ALICE L WALTON SCHOOL OF MEDICINE

Obstructive Congenital Heart Defects

Last updated January 13, 2026 · 17 min read · ScholarRx

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Adapted from: Acyanotic Obstructive Congenital Heart Defects

Listen to this Brick

0:00 / 26:55 1x

Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of obstructive congenital heart defects.

A coarctation is a stricture, or narrowing, of a canal or blood vessel; in this case, it is the narrowing of the lumen of the aorta (Figure 1).

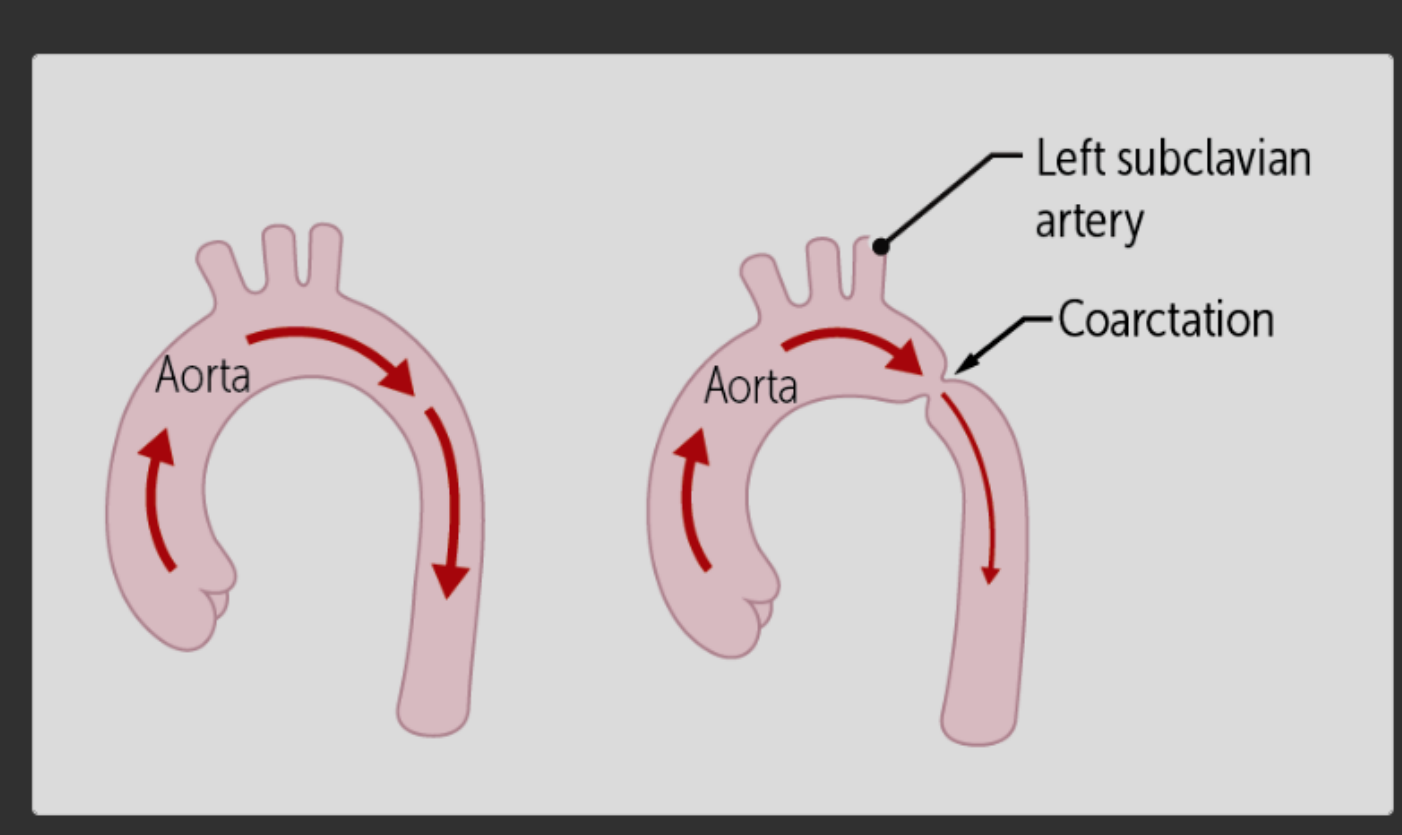


Figure 1 Normal aorta vs coarctation of the aorta

It's most often located in the descending aorta just distal to the left subclavian artery and near the attachment of the ligamentum arteriosum, the remnant of the fetal ductus arteriosus.

The congenital form of coarctation of the aorta (CoA) is much more common than the adult form and accounts for ~5% of CHDs. It occurs in 0.3% of live births and is twice as common in males as in females.

chest pain with exertion.

CoA is associated with several other CHDs including bicuspid aortic valve, ventricular septal defect, and patent ductus arteriosus. CoA is also present in 10% to 15% of patients with Turner syndrome.

CLINICAL CORRELATION

Turner syndrome is a chromosomal abnormality in which a female has only one X chromosome (45, XO) and presents with findings such as short stature, lymphatic lesions (ie, webbed neck, lymphedema in feet and hands), and a horseshoe kidney.

How does the severity of CoA differ between patients presenting shortly after birth and those presenting during childhood?

exact mechanism of either type is not fully understood.

Acquired CoAs are rare, but they arise in adulthood because of severe atherosclerosis or inflammatory diseases of the aorta, such as Takayasu arteritis.

From what embryonic structure does the arch of the aorta arise?

Physiology. Because CoA is an obstructive lesion, the main hemodynamic issue is increased resistance to blood flow through the narrowed segment of the descending aorta. This is problematic because tissues distal to the coarctation require oxygen-rich blood to survive. So, the body adapts in at least one of two ways to get blood past the narrowed aorta:

- By increasing systemic blood pressure: higher blood pressures proximal to the aortic narrowing will increase arterial pressure proximal to the stenosis (eg, in the arms a ... ng cause of

What is the most common presentation of children with CoA?

Radiology. Echocardiography with Doppler is the imaging modality of choice, as it can be used to detect the aortic narrowing and estimate the severity.

Chest X-Ray. Chest x-ray is not required for diagnosis, but, in adults with uncorrected CoA, it may reveal "notching" of the inferior surface of the ribs, mostly noted from the third to the eighth ribs (Figure 4).

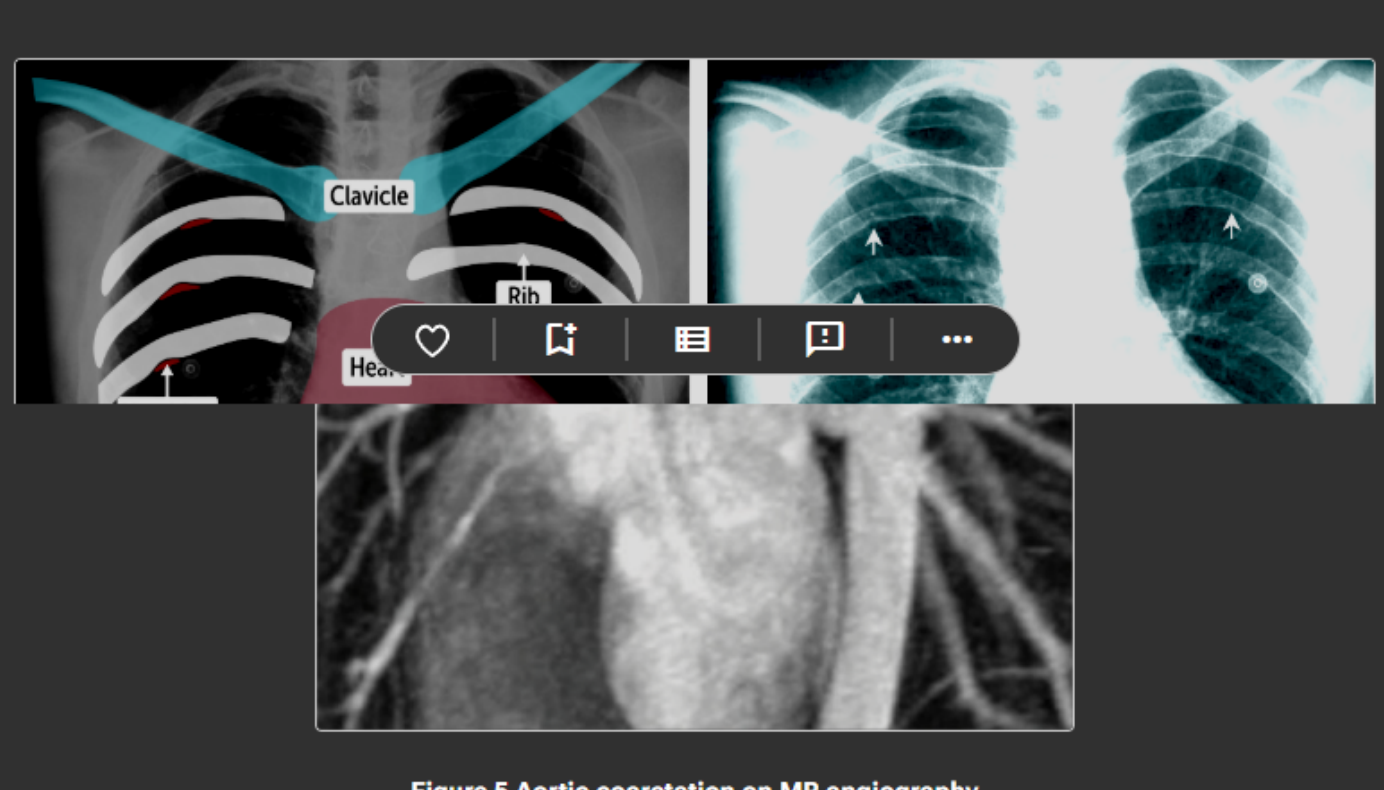


Figure 5 Aortic coarctation on MR angiography

What are the imaging modalities used to confirm a diagnosis of CoA?

Management

Management depends on the severity of the coarctation, patient's age, and clinical presentation.

Neonates and Young Children. Surgical interventions include resection with

What is the preferred method of treatment of CoA in older children and adults?

What Is Congenital Pulmonary Valve Stenosis? (LO#1)

Pulmonary valve stenosis, also called pulmonary stenosis and pulmonary artery stenosis, is the third most common type of CHD, accounting for ~8% of CHD cases. A thickened pulmonary valve narrows the artery's lumen and limits blood flow into the pulmonary artery and lungs.

Clinical Presentation

The symptoms of pulmonary valve stenosis depend on the degree of narrowing (obstruction). Mild stenosis is typically asymptomatic, and most patients with this defect have mild disease. Severe stenosis, however, causes a drastic decrease in blood flow to the lungs (and therefore in oxygenation), leading to syncope, chest pain, and dyspnea on exertion (all due to hypoxemia).

Pathophysiology

In congenital pulmor ... pulmonary valve to thicken and the commissures to fuse (Figure 7).

What are common complications arising from worsening pulmonary valve stenosis?

CLINICAL CORRELATION

When the valve is entirely atretic (pulmonary atresia- absent or severely underdeveloped valve), there is no communication between the right ventricle and the lungs. In such cases, the anomaly is associated with a hypoplastic right ventricle and an ASD; blood reaches the lungs through a patent ductus arteriosus.

Diagnosis

Pulmonary valve stenosis is diagnosed by physical exam and radiology studies.

Physical Examination. Most patients present in early childhood with a harsh systolic ejection murmur. The murmur often radiates to the back. Electrocardiography ... and RVH.

valve and/or adjacent aortic wall obstruct the lumen and reduce blood flow out of the left heart into the peripheral circulation.

CLINICAL CORRELATION

Congenital aortic valve stenosis is much less common than the adult form, which is usually due to calcification of the valve with aging. Think of a congenital cause of AVS when the findings appear before age 50 years.

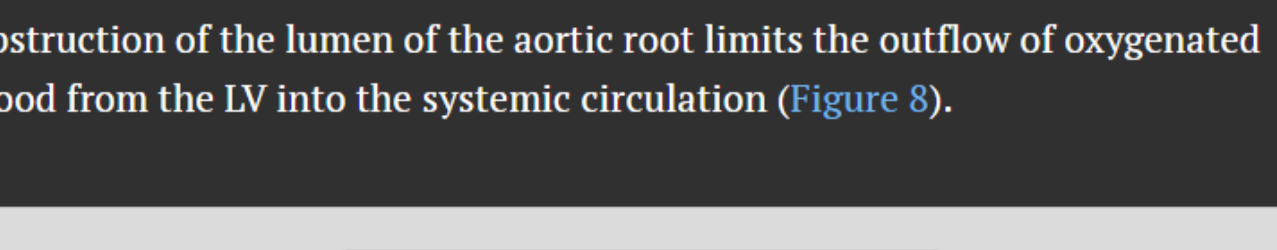
Clinical Presentation

Most children with AVS are asymptomatic since they have only mild or moderate stenosis. With increasing stenosis, patients will develop dyspnea (due to heart failure), as well as dizziness, syncope, and chest pain on exertion due to reduced arterial blood flow.

In the neonatal period, critical AVS may present as shock, whereas in childhood, severe AVS more typically presents with symptoms of heart failure.

Pathophysiology

Obstruction of the lumen of the aortic root limits the outflow of oxygenated blood from the LV into the systemic circulation (Figure 8).



ALICE L WALTON SCHOOL OF MEDICINE

Cyanotic Congenital Heart Defects (Right-to-Left Shunts)

Last updated February 28, 2026 · 22 min read · ScholarRx

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Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of cyanotic congenital heart defects.

What do you want to learn?

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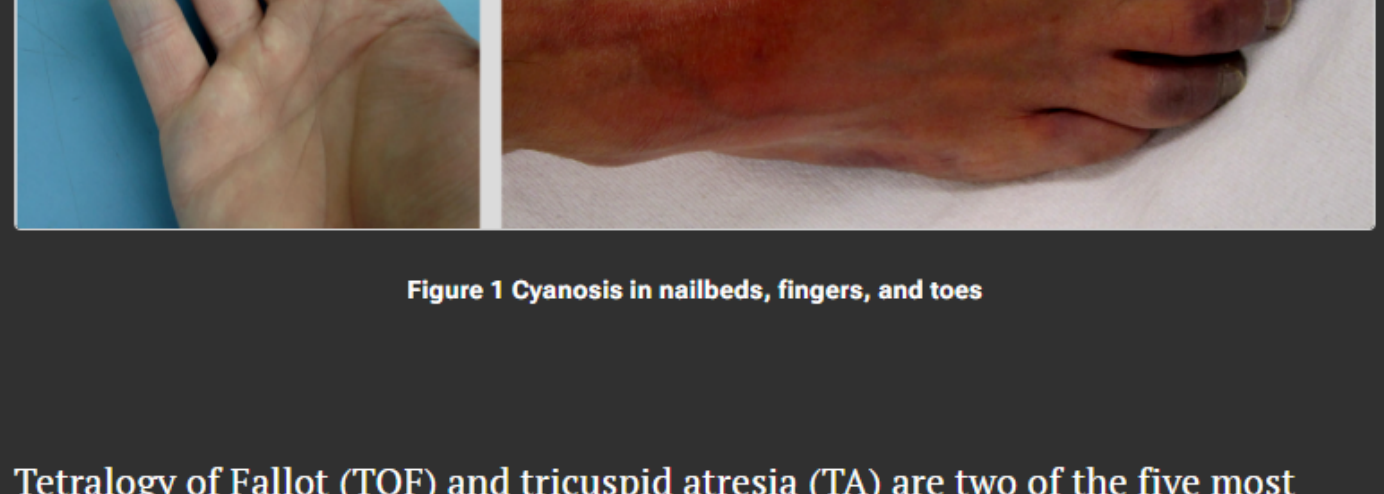


Figure 1 Cyanosis in nailbeds, fingers, and toes

Tetralogy of Fallot (TOF) and tricuspid atresia (TA) are two of the five most common cyanotic CHDs. They both also include **obstructive defects on the right side of the heart.** These obstructions limit blood flow into the pulmonary vasculature.

The other three common cyanotic CHDs are considered **"mixing defects,"** which are a subset of cyanotic CHDs where oxygen-rich and oxygen-poor blood mix, resulting in reduced oxygen delivery to the body and cyanosis. These CHDs are: truncus arteriosus, transposition of the great arteries, and total anomalous pulmonary venous return.

What Is Tetralogy of Fallot? (LO#1)

Tetralogy of Fallot is the name for the combination of four ("tetra") specific CHDs (Figure 2):

- Pulmonary valve stenosis
- Right ventricular (RV) hypertrophy (due to the valve stenosis)
- Ventricular septal ...
- Overriding aorta, in which the aorta is displaced to the right and receives

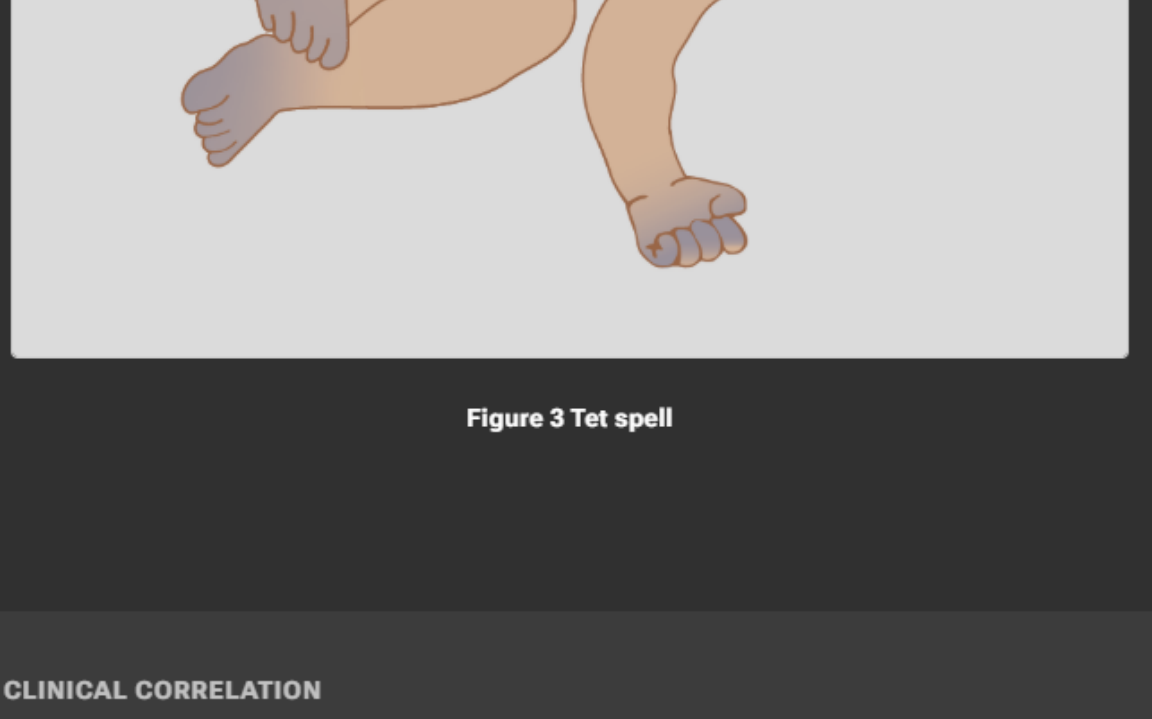


Figure 3 Tet spell

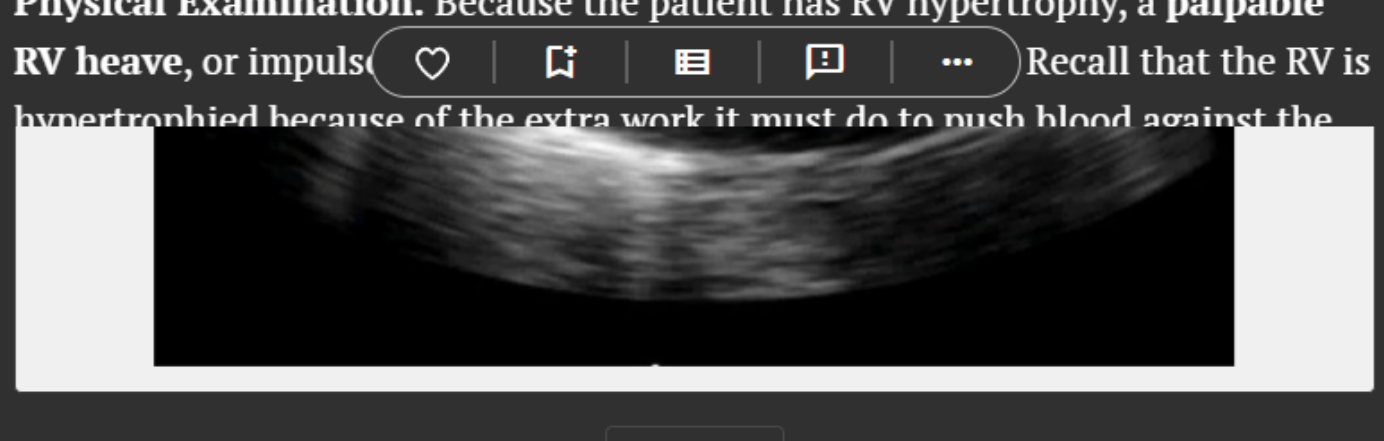
CLINICAL CORRELATION

Interestingly, squatting helps the symptoms of a tet spell because this action increases systemic vascular (arterial) resistance, which decreases the amount of deoxygenated blood crossing over the VSD.

Diagnosis

A few physical examination features can be found in patients with TOF, but diagnosis is best done with echocardiography.

Physical Examination. Because the patient has RV hypertrophy, a **palpable RV heave,** or impuls ... Recall that the RV is hypertrophied because of the extra work it must do to push blood against the



QUIZ

Figure 6 Echocardiography in tetralogy of Fallot

Management

Treatment can be divided into acute symptom management and surgery.

Managing Tet Spells. The first step in caring for a child during a tet spell is to focus on symptom relief. One method of relief is to have a parent hold the child with the child's knees toward their chest (see Figure 3).

Oxygen supplementation will be effective, as it will dilate the pulmonary arteries, which allows the blood from the RV to flow more freely into the lungs. Morphine is sometimes used during tet spells for sedation and pain relief. Propranolol is also used to relax the obstruction and promote pulmonary blood flow.

Surgery. Surgical repair of TOF is generally done in the first year of life. It involves opening the pulmonary stenosis by either open surgical or catheter-based balloon valvuloplasty and, at the same time, closing the VSD. This relieves the right-to-left shunting and cyanosis. Patients tend to do well after surgery, although arr ... gitation are